

Digital Forensic Analysis of a Camera Memory Card Image

Practice case report | Autopsy | Case001_CanonCard

1. Case Details

Examiner	Lucky Bhati
Analysis date	02 October 2026
Case name	Case001_CanonCard
Purpose	Practice analysis using a public training image (no real person's data)

2. Objective

To examine a forensic image of a digital camera memory card, identify the device and photo metadata (EXIF), and find any deleted image files and whether their content can be recovered.

3. Tools Used

- Autopsy (open-source digital forensics platform) for analysis, EXIF parsing and deleted-file review
- Windows PowerShell (Get-FileHash) for evidence hashing
- Ingest modules: File Type Identification, Extension Mismatch Detector, Hash Lookup, Exif Parser, Embedded File Extractor

4. Evidence Details

Evidence file	nps-2009-canon2-gen3.E01 (EnCase E01 disk image)
Source	Digital Corpora (public forensic practice corpus), nps-2009-canon2
Hash (MD5)	9B1F461C0F16A9703DAC74B327A54694
Hash recorded	After download, before analysis (PowerShell Get-FileHash)
Device model (from EXIF)	Canon PowerShot SD800 IS

5. Methodology

- Downloaded the image and calculated its MD5 hash before opening it.
- Created a new case in Autopsy and added the .E01 file as a Disk Image data source.
- Ran the ingest modules listed above.
- Reviewed **Analysis Results > EXIF Metadata** and **File Views > Deleted Files**.
- Checked file Metadata and preview for each deleted item, and recorded names, paths and sizes.

6. Findings

6.1 Photos and EXIF metadata. Autopsy reported 33 images with EXIF metadata. The device model is Canon PowerShot SD800 IS. The photos were created on 23 Dec 2008, between 14:12:38 IST (earliest) and 14:26:13 IST (latest), as displayed by Autopsy.

6.2 Deleted files. Five deleted image entries were found:

#	File name	Path	Size (bytes)	Preview
1	_MG_0037.JPG	/vol_vol2/DCIM/100CANON/	1,347,778	Not available

#	File name	Path	Size (bytes)	Preview
2	_MG_0025.JPG	/vol_vol2/DCIM/100CANON/	791,333	Not available
3	_MG_0030.JPG	/vol_vol2/DCIM/100CANON/	867,833	Not available
4	_MG_0035.JPG	/vol_vol2/DCIM/100CANON/	820,105	Not available
5	f0000000.jpg	/vol_vol2/\$CarvedFiles/1/	855,935	Displayed

Files 1 to 4 are deleted entries in the camera's DCIM folder. File 5 is located under *\$CarvedFiles*, which means Autopsy found it by file carving (searching the raw data for JPEG signatures) and not through the normal file table. Its content could be previewed, so one image was recovered (Figure 2).

6.3 Content of the recovered image. The recovered image is a photograph of a computer monitor. A TextEdit window is open and shows a large digit "1". Part of a Terminal window is also visible behind it, and the screen clock shows 1:13 PM. This appears to be a staged test photo and not personal data.

7. Conclusion

The image contains 33 photos taken with a Canon PowerShot SD800 IS on 23 December 2008. Five deleted image entries were identified. The content of one deleted image (f0000000.jpg) was recovered and previewed, while the content of the other four could not be previewed in Autopsy.

8. Limitations

- Only Autopsy was used. The four non-previewable files may be recoverable with other tools or may have been partly overwritten; this was not tested.
- Dates are shown as displayed by Autopsy using the India time zone setting. Cameras store local time without a time zone, so the real time zone of the photos is unknown. The on-screen clock in the recovered photo (1:13 PM) also differs from the EXIF time, so the camera clock may not be accurate.
- The hash was not compared with a published value.

9. Chain of Custody Note

Date	Action	By
Oct 2026	Downloaded evidence image from Digital Corpora and calculated MD5 hash	Lucky Bhati
Oct 2026	Opened image in Autopsy (read-only analysis, original file not modified)	Lucky Bhati
02 Oct 2026	Generated Autopsy HTML report and saved a copy of the recovered file	Lucky Bhati

10. Evidence Screenshots

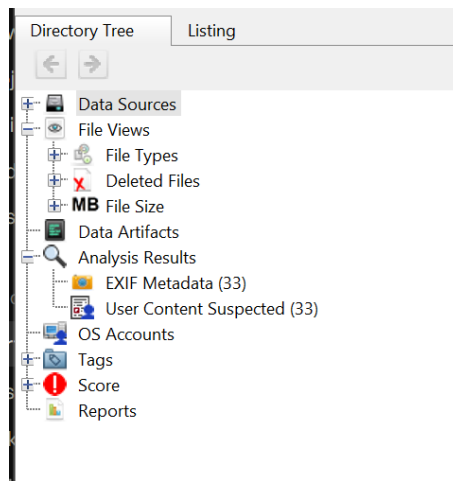


Figure 1: Autopsy directory tree after ingest: 33 EXIF Metadata results and Deleted Files view



Figure 2: Recovered image f0000000.jpg (carved from unallocated space)